DNA, chromosomes and genes

BACKGROUND

nucleus (say 'nu-kle-us') – the round structure at the centre of a cell, containing genetic information that controls what the cell does

A new life begins as just one **cell**. All the information for growing a living thing is found inside the nucleus of this cell. For most animals, this cell is a fertilised egg, containing **genetic information** from two parents. This genetic information is stored in DNA, in genes and chromosomes.

DNA

- The letters DNA stand for deoxyribonucleic acid (*say* de-ox-ee-rye-bo-new-clee-ick ass-id). When people talk about DNA, they usually mean the information that DNA contains, not the acid itself.
- DNA is found inside the cells of nearly all living things.
- DNA is like a set of instructions for building and operating a living thing. It provides the information to make the **proteins** that make up cells, as well as instructions as to how the cells form, what shape they will be, what things they can do and so on.

Chromosomes and genes

- Chromosomes are tiny threads made of DNA. Genes are short pieces of these threads.
- Each chromosome consists of thousands of different genes.
- Every gene carries information about a specific characteristic.
- Chromosomes are found inside the **nucleus** of cells. In each cell, the genetic information is exactly the same, containing the same instructions.
- Chromosomes come in pairs. Humans have 23 pairs of chromosomes.
- One chromosome of every pair comes from the mother and one in every pair comes from the father.

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ISBN 978 1 92571 064 9

DNA, chromosomes and genes CONT'D

☞ **1** Are the following statements true or false?

a) The letters DNA stand for dioxide ribbons and nuclear acids. True False

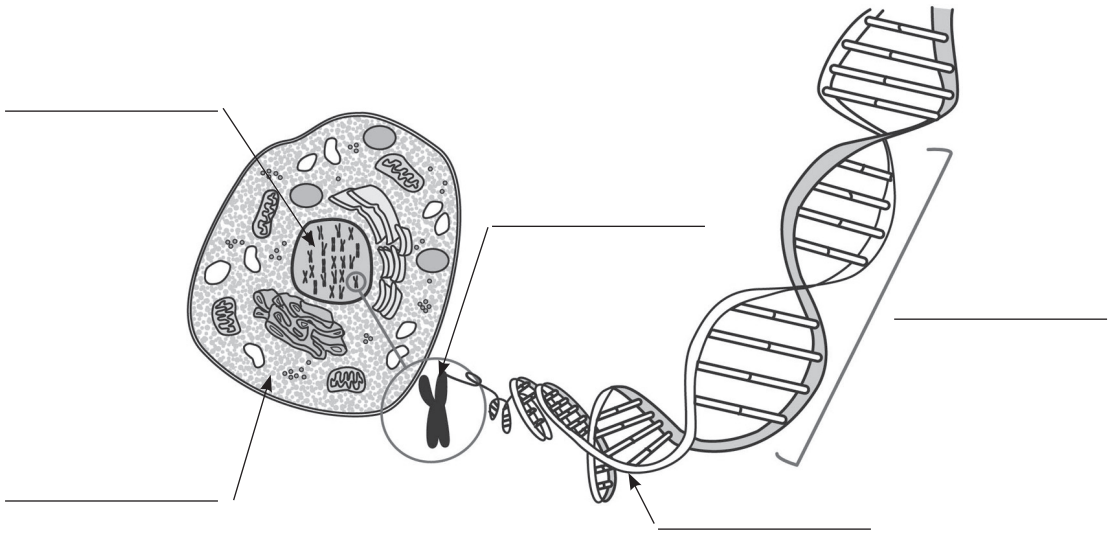
b) DNA contains information to make proteins in particular ways. True False

c) DNA is the genetic material of living things. True False

d) Our parents can decide on and control what material goes into our genes. True False

2 Look at the diagram. Label the features listed in the box.

chromosome	nucleus	cell	gene	DNA



3 Rearrange these sentence parts to make sentences about chromosomes. Write the correct sentences on the lines provided.

a) tiny threads DNA Chromosomes are that are made of

_____

_____.

b) the genetic material the same instructions contains exactly In each cell,

_____

_____.

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